Project No. 项目编号	TXW686013B0
Customer 客户名称	
Module No. 客户型号	
Product type **产品内容**	**Standard LCD Module** **TFT：480*RGBx1280Dots** **6.86”TFT LCD**

客户确认Customer Approval	
项目负责人Project Manager	
品质主管Director of Quality	
采购工程师Purchasing Engineer	

PREPARED BY	CHECKED BY	APPROVED BY

Shenzhen Tianxianwei Technology Co., Ltd.

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1. Introduction

1.1 Scope of application

This specification applies to the LCD module that is supplied by TIAN XIAN WEI TECHNOLOGY CO., LTD. This LCD module should be designed for mobile phone use.
LCD specification: Dots 480xRGBx1280.
As to basic specification of the driver IC, refer to the IC (NV3052C) specification and data book.
All material & processing of the LCD module should be Lead Free.

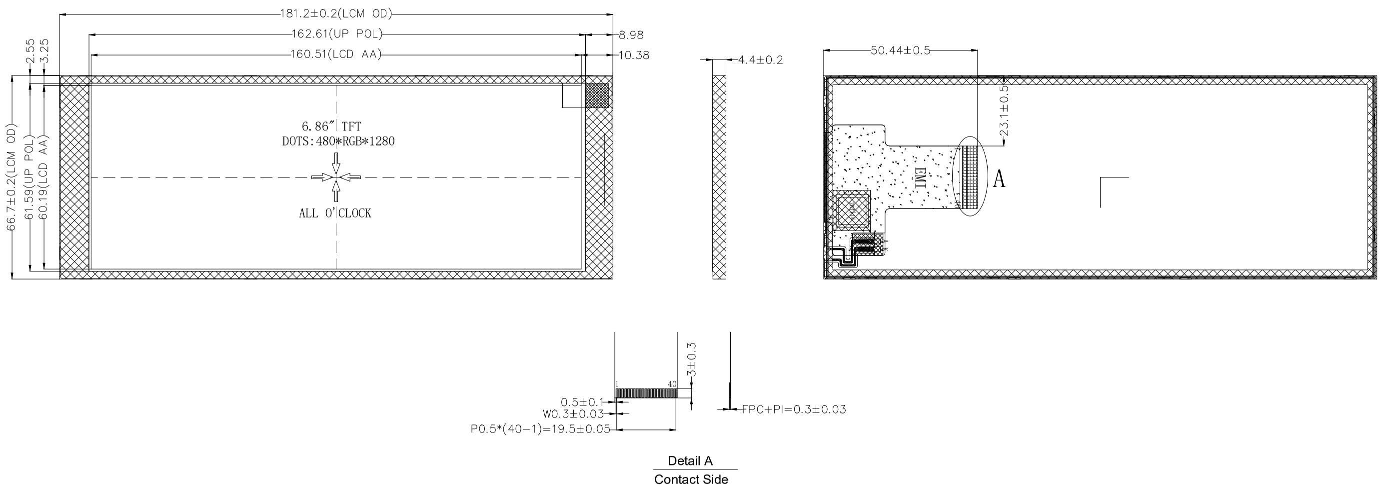
1.2 TFT features:

Structure: TFT PANNEL+IC +FPC+BL;
ALL O'CLOCK Type LCD
480 dot-segment and 1280 dot-common outputs;
16.7M Color can be selected by software;
White LED back light;
3line SPI or 16/24bit RGB interface

2. LCM General specification

ITEM	Sandard value	Unit
LCD Type	Normally Black	--
Drive element	TFT active matrix	--
Number of pixels	480*3RGB(H)X1280(V)	Dots
Pixel arrangement	RGB stripe	--
Pixel Pitch (W*H)	0.1254(W)x 0.1254(H)	mm
Active area	60.19(H) ×160.51 (V)	mm
Viewing direction	ALL O'CLOCK	--
TFT Driver IC	NV3052C	--
TFT interface	3line SPI or 16/24bit RGB Interface	--
Approx. Weight	TBD	g
LCM Size(W*H*T)	66.70(W) x181.20(H) x4.40(T)	mm
Touch structure	--	--
Touch Driver IC	--	--
Touch Interface	--	--

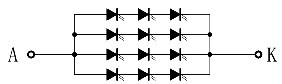
REV	DESCRIPTION	DRG DATE	DRAWN BY
	First Design	2023. 08. 31	



Pad No.	Pad Name
1	LED-
2	LED+
3	GND
4	VDD
5	R0/NC
6	R1/NC
7	R2/NC
8	R3
9	R4
10	R5
11	R6
12	R7
13	G0/NC
14	G1/NC
15	G2
16	G3
17	G4
18	G5
19	G6
20	G7
21	B0/NC
22	B1/NC
23	B2/NC
24	B3
25	B4
26	B5
27	B6
28	B7
29	GND
30	PCLK
31	DISP
32	HSYNC
33	VSYNC
34	DEN
35	NC
36	GND
37	SCL
38	CS
39	RESET
40	SDA

NOTES:
1. DISPLAY TYPE: 6.86", 480*1280 TFT LCD
2. DISPLAY MODE: transmissive Normally Black
3. VIEWING DIRECTION: ALL
4. DRIVER IC:NV3052C
5. LCM (White 9 AVG 1/9) :
   Brightness: TBDcd/m2
   Uniformity:80%(MIN)
6. BACK LIGHT: 12 chip white LEDs If=160mA,Vf=8.4-9.6V
7. OPERATING TEMP: -20° C TO 70° C,STORAGE TEMP: -20° C TO 80° C
8. * Critical Parameter,( ) ref Parameter,[ ] cpk Parameter
   Unspecified Tolerances:±0.20mm
   Modification mark:
9. SUGGESTION:TP window size unilateral increase 0.3~0.5mm than LCM A.A
10. REQUIREMENTS ENVIRONMENTAL PROTECTION: RoHS

LED CIRCUIT DIAGRAM



深圳市天显威科技有限公司 Shenzhen Tian xian wei Technology Co., Ltd.				Mobile: 0755-82597676	
PART NO.:	TXW686013B0			MODULE NO.:	00
DRAWN BY:	LIU	DATE:	2023.08.31	PROJECTION:	
CHECKED BY:		DATE:		SCALE:N.T.S	UNIT: mm
APPROVED BY:		DATE:		SHEET: 1 OF 1	

3.Absolute Maximum Rating

Characteristics	Symbol	Min.	Max.	Unit
LCM Operating Temperature	T_{OPR}	-20	+70	°C
LCM Storage Temperature	T_{STG}	-20	+80	°C
Humidity	RH	--	90	%

4.Electrical Characteristics

4.1 TFT DC Characteristics

Characteristics	Symbol	Min.	Typ.	Max.	Unit
Supply Voltage for(DC/DC)	VDD	2.5	3.3	3.3	V

4.2 Back-Light Unit Characeristics

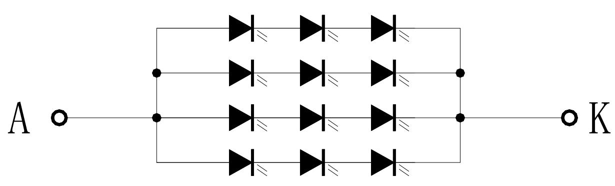
The back-light system is an edge-lighting type with 12 white LEDs. The characteristics of the back-light are shown in the following tables.

Characteristics	Symbol	Min.	Type	Max.	Unit	Notes
Forward Voltage	V_F	8.4	--	9.6	V	--
Forward current	I_F	--	160	--	mA	--
Luminance(With LCD)	Lv	--	TBD	--	cd/m²	--
LED life time	N/A	--	30,000	--	Hr	Note 1

Note:

(1) The "LED life time" is defined as the module brightness decrease to 50% of original brightness at I_L=40mA/LED. The LED life time could be decreased if operating I_L is larger than 45mA/LED.

Backlight circuit diagram shown in below:



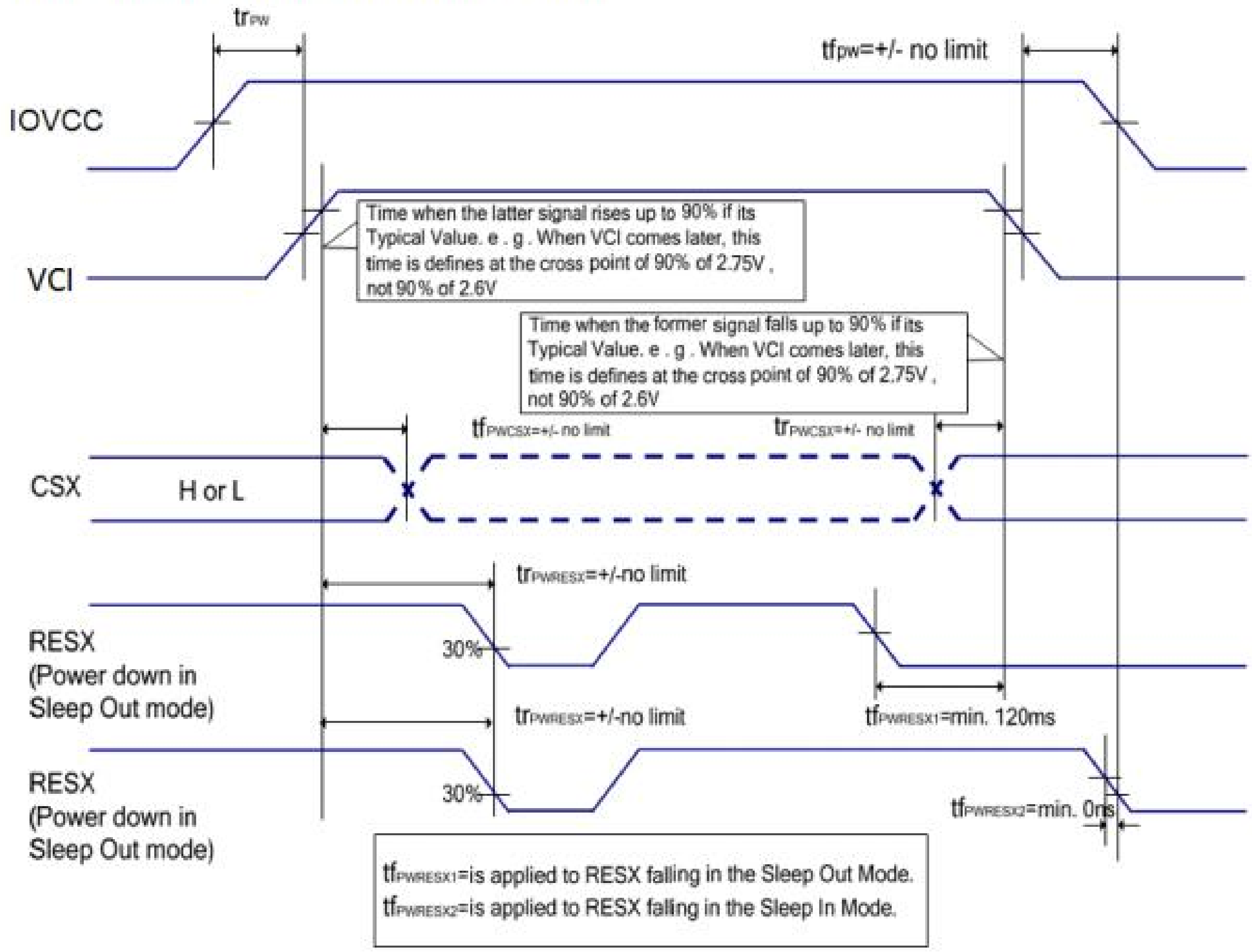
5. Module Function Description

Pin No.	Symbol	LCM Description	remarks
1	LEDK	POWER SUPPLY+ FOR BACKLIGHT ANODE	
2	LEDA	POWER SUPPLY- FOR BACKLIGHT CATHODE	
3	GND	Power ground	
4	VDD	Power Supply For LCD.	
5	R0/NC	Red data/No Connection.	
6	R1/NC	Red data/No Connection.	
7	R2/NC	Red data/No Connection.	
8	R3	Red data	
9	R4	Red data	
10	R5	Red data	
11	R6	Red data	
12	R7	Red data	
13	G0/NC	Green data/No Connection.	
14	G1/NC	Green data/No Connection.	
15	G2	Green data	
16	G3	Green data	
17	G4	Green data	
18	G5	Green data	
19	G6	Green data	
20	G7	Green data	
21	B0/NC	Blue data/No Connection.	
22	B1/NC	Blue data/No Connection.	
23	B2/NC	Blue data/No Connection.	
24	B3	Blue data	
25	B4	Blue data	
26	B5	Blue data	
27	B6	Blue data	
28	B7	Blue data	
29	GND	Power ground	
30	PCLK	Dot clock signal for RGB interface operation.	
31	DISP	No Connection.	
32	HS	Horizontal sync. Signal in RGB I/F	
33	VS	Vertical sync. Signal in RGB I/F	
34	DEN	Data enable signal in RGB I/F mode 1	
35	NC	No Connection.	
36	GND	Power ground	
37	SCL	Serial clock signal.	
38	CS	Chip selection signal.	
39	RESET	Global reser pin	
40	SDA	Serial data input/output pin.	

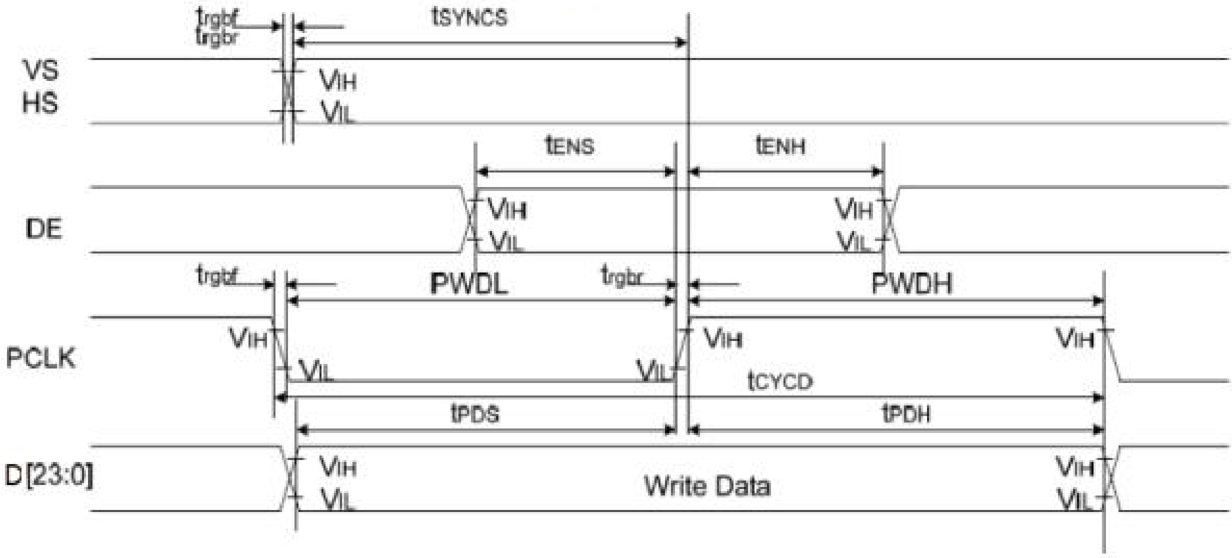
6. Timing Characteristics

Case 1 – RESX line is held high or unstable by host at power on

If RESX line is held High or unstable by the host during Power On, then a Hardware Reset must be applied after both VCI and IOVCC have been applied – otherwise correct functionality is not guaranteed. There is no timing restriction upon this hardware reset.



Parallel 24/18/16-bit RGB Interface Timing Characteristics



Signal	Symbol	Parameter	min	max	Unit	Description
VS/HS	t_{SYNCS}	VS/HS setup time	5	-	ns	24/18/16-bit bus RGB interface mode
	t_{SYNCH}	VS/HS hold time	5	-	ns	
DE	t_{ENS}	DE setup time	5	-	ns	
	t_{ENH}	DE hold time	5	-	ns	
D[23:0]	t_{POS}	Data setup time	5	-	ns	
	t_{PDH}	Data hold time	5	-	ns	
PCLK	PWDH	PCLK high-level period	13	-	ns	
	PWDL	PCLK low-level period	13	-	ns	
	t_{CYCD}	PCLK cycle time	28	-	ns	
	t_{rgbr},t_{rgbf}	PCLK,HS,VS rise/fall time	-	15	ns	

Note 1: IOVCC=1.65 to 3.6V, VCI=2.5 to 6V, VSSA=VSS=0V, Ta=-30 to 70℃

Serial interface characteristics (SPI)

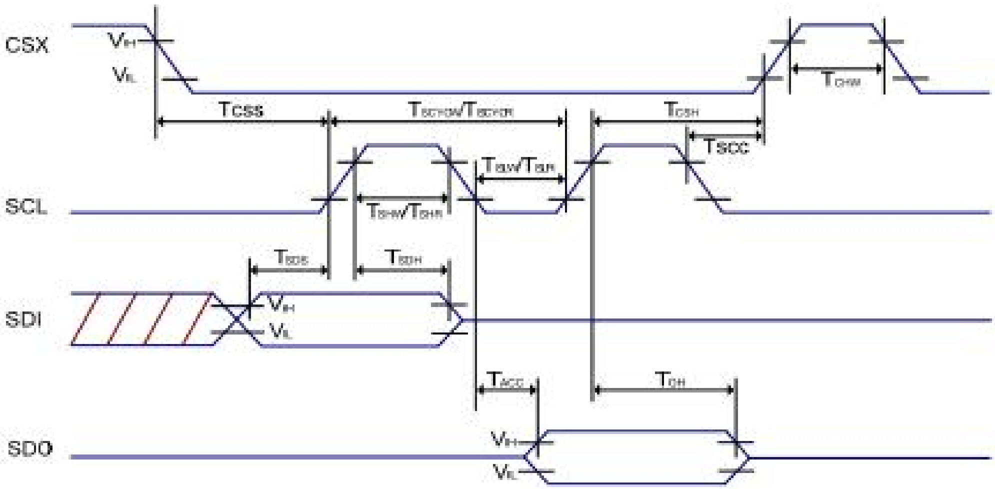


Figure: 3-pin Serial Interface Characteristics

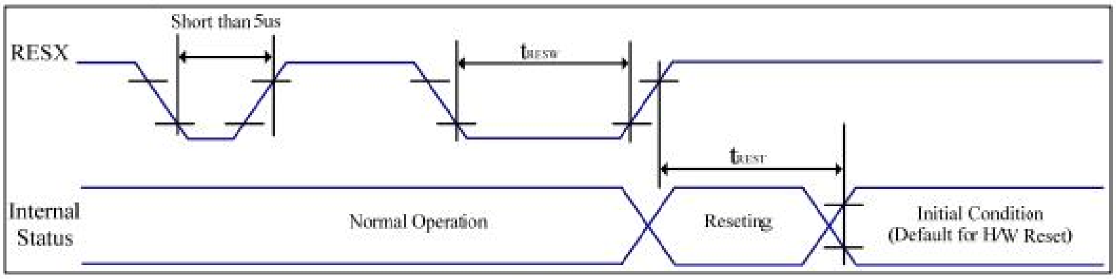
Table: SPI Interface Characteristics

Signal	Symbol	Parameter	MIN	MAX	Unit	Description
CSX	T_{CSS}	Chip select setup time	15	-	ns	-
	T_{CSH}	Chip select hold time	15	-	ns	
	T_{SCC}	Chip select setup time	20	-	ns	
	T_{CHW}	Chip "H" pulse width	40	-	ns	
SCL	T_{SCYCW}	Serial clock cycle (Write)	66	-	ns	-
	T_{SHW}	SCL "H" pulse width (Write)	10	-	ns	
	T_{SLW}	SCL "L" pulse width (Write)	10	-	ns	
	T_{SCYCR}	Serial clock cycle (Read)	150	-	ns	-
	T_{SHR}	SCL "H" pulse width (Read)	60	-	ns	
	T_{SLR}	SCL "L" pulse width (Read)	60	-	ns	
SDI	TSDS	Data setup time	10	-	ns	-
	T_{SDH}	Data hold time	10	-	ns	
	T_{ACC}	Access time	10	50	ns	For maximum C_L=30pF For minimum C_L=8pF
	T_{OH}	Output disable time	15	50	ns	

Note 1: IOVCC=1.65 to 3.6V, VCI=2.5 to 6V, VSSA=VSS=0V, Ta=-30 to 70℃

Note 2: The rise time and fall time (tr, tf) of input signal is specified at 15 ns or less. Logic high and low levels are specified as 30% and 70% of IOVCC for Input signals.

Reset timing characteristics



VSS=0V, IOVCC=1.65V to 3.6V, VCI=2.5V to 6.0V,Ta = -30℃ to 70℃

Symbol	Parameter	Related Pins	MIN	TYP	MAX	Note	Unit
t_{RESW}	*1) Reset low pulse width	RESX	10	-	-	-	us
t_{REST}	*2) Reset complete time	-	-	-	5	When reset applied during Sleep in mode	ms
		-	-	-	120	When reset applied during Sleep out mode	ms

Table: Reset input timing

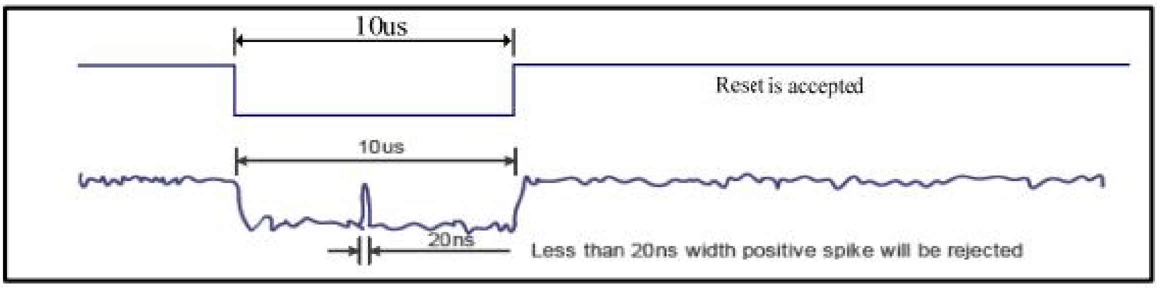
Note 1: Due to an electrostatic discharge on RESX line,spike does not cause irregular system reset according to the table below.

RESX Pulse	Action
Shorter than 5us	Reset Rejected
Longer than 10us	Reset
Between 5us and 10us	Reset starts (It depends on voltage and temperature condition.)

Note 2: During the resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120ms, when Reset Starts in Sleep Out mode. The display remains the blank state in Sleep In mode) , then return to default condition for H/W reset.

Note 3: During Reset Complete Time, ID1/ID2/ID3 and VCOM value in OTP will be latched to internal register. After a rising edge of RESX, there is a H/W reset complete time (Trest) which lasted 5ms..The loading operation will be done every time during this reset.

Note 4: Spike Rejection also applies during a valid reset pulse as shown below:



Note 5. It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120 msec.

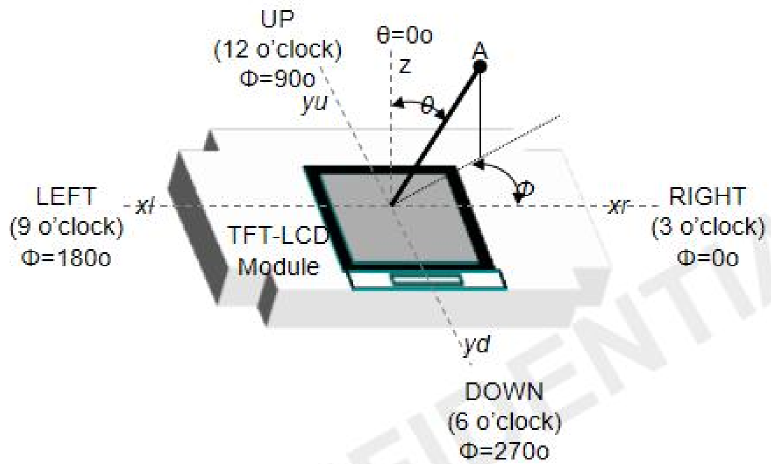
7.Optical Characteristics

<Table 5. Optical Specifications> [Ta=25±2℃]

Parameter		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
Viewing Angle Range	Horizontal	Θ_3	CR > 10	75	85	-	Deg.	Note 4.1
		Θ_9		75	85	-	Deg.	
	Vertical	Θ_{12}		75	85	-	Deg.	
		Θ_6		75	85	-	Deg.	
Contrast Ratio		CR	Θ = 0°	1000	1500	-		APF Note 4.2/4.3
Cell Transmittance		Tr		3.74	4.4	-	%	
Reproduction of color		Rx	Θ = 0°					With BLU @C Light Note 4.4
		Ry						
		Gx						
		Gy						
		Bx						
		By						
		Wx		0.263	0.293	0.323		
		Wy		0.292	0.322	0.352		
Color Gamut			Θ = 0°	65	70	-	%	
Response Time		Tr	Ta= 25°C Θ = 0°	-	16	21	ms	Note 4.5
		Tf		-	14	19	ms	
		Tgray		-	30	40	ms	

Note 1: Viewing angle is the angle at which the contrast ratio is greater than 10. The viewing angles are determined for the horizontal or 3, 9 o'clock direction and the vertical or 6, 12 o'clock direction with respect to the optical axis which is normal to the LCD surface (see Figure 8).

<Figure 7. Viewing Angle Range Is Defined As Follows>



Note 2: Contrast measurements shall be made at viewing angle of $\Theta=0^{\circ}$ and at the center of the LCD surface. Luminance shall be measured with all pixels in the view field set first to white, then to the dark (black) state. Luminance Contrast Ratio (CR) is defined mathematically.

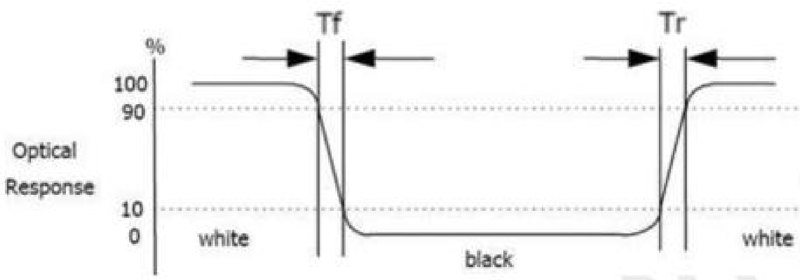
$$CR = \frac{\text{Luminance when displaying a white raster}}{\text{Luminance when displaying a black raster}}$$

Note 3: Transmittance is the Value with Polarizer.

Note 4: The color chromaticity coordinates specified in Table 16 shall be calculated from the spectral data measured with all pixels first in red, green, blue and white. Measurements shall be made at the center of the panel.

Note 5: The electro-optical response time measurements shall be made as Figure 9 by switching the "data" input signal ON and OFF. The times needed for the luminance to change from 10% to 90% is Tr, and 90% to 10% is Tf.

<Figure 8. Response Time Testing>



8. Reliability Test Item

No.	Test Item	Test Condition	Notes
1	High Temp. Storage	+80°C / 96H	1. Functional test isOK. Missing Segment,short, unclear segment non-display,display abnormally and liquid crystal leakare un-allowed. 2. No low temperature bubbles,end seal loose andfall, frame rainbow.
2	Low Temp. Storage	-20°C / 96H	
3	High Tempe. Operating	+70°C / 96H	
4	Low Tempe. Operating	-20°C / 96H	
5	High Temperature /Humidity storage	60°C x 90%RH /96H	
6	Thermal and cold shock	Static state，-20℃（30min）~70℃（30min），10 cycles	

9. Packing Method----TBD

- END -